From AI Strategy to University Rules in the GCC

***Abstract*—How can readers tell which official AI-governance documents govern a university activity, rather than merely express national ambition? We present the Four-Layer Public-Evidence Audit (4L-PEA) for six Gulf Cooperation Council (GCC) jurisdictions: national direction (L1), cross-sector controls (L2), higher-education-applicable controls (L3), and institutional rules (L4). Dated official records are coded explicit (E), partial (P), or unlocated (U) under layer-specific gates. In a retrospectively recovered 36-record source register with a 23 July 2026 document-date cutoff, L3=E in one jurisdiction and L4=E in six; a selected 57-institution panel yields 20 E, 12 P and 25 U. The study makes the unequal L3/L4 thresholds explicit. A worked UAE example, linked official clauses, a four-step workflow and editable audit forms make the protocol reusable. Two distinct Omani institutional sources show why deleting one anchor from the 36-record register is not equivalent to withdrawing all L4 evidence in a separate institution panel. The interval representation of P is descriptive, not a measure of undiscovered sources or coder accuracy. Exact historical website visibility, the complete original eligibility universe, and independent intercoder reliability are not established. The contribution is a transparent, bounded documentary audit, not a claim of policy transmission or implementation.**

***Index Terms*—AI governance, GCC, higher education, bilingual document audit, public traceability, evidence admissibility, policy reuse.**

I. Why This Audit Is Needed

An instructor needs to know whether AI use is permitted in an assignment, what must be disclosed, and who verifies a student's work. A national strategy can express ambition without answering those questions. The practical research question is therefore narrower than "Which country has the best AI governance?": *Which official documents can a reader actually locate and check at each governance level?*

Prior studies have compared GCC national strategies [1] and university AI policies across 40 universities in six global regions [2], 116 US research institutions [3], and 343 institutions in five countries [4]. Work on assessment [5], university policy development [6], and systematic or institution-level governance [7], [8], [9], [10], [11] illustrates substantial policy variety. Multi-level governance and policy translation are distinct concepts [12], [13]. Our contribution is the Four-Layer Public-Evidence Audit (4L-PEA), a *source-to-claim procedure*: it shows how to record the evidence for four levels without assuming that a policy at one level caused, authorized, or was implemented by another.

A reader example. Suppose a ministry advertises AI innovation, while a university publishes a formal rule specifying allowed assistance in assessments. The ministry announcement can document ambition but cannot tell the student what that university permits. Our audit asks two different questions: *does a higher-education (HE)-applicable, operative instrument exist at the sector level?* and *does a university-issued rule exist at a named institution?* It records direct textual evidence for each answer. It never assumes that one document caused the other or that staff comply with the document simply because it is online.

The cutoff applies to the document corpus, not to the publication dates of the 2026 scholarship used for context. The paper answers three questions: (1) Which layers contain a qualifying public instrument? (2) Does the difference between sector-level HE evidence and institutional rules persist under stated evidence and sampling changes? (3) What minimum record would enable another researcher to check each classification? The accompanying material gives that reader a workflow, not a new country score.

II. The Four-Layer Decision Rule

A. What E, P and U mean

For each GCC jurisdiction j and layer ℓ, record one nominal state $L_{\ell j} \in \{\mathsf{E}, \mathsf{P}, \mathsf{U}\}$. **E (explicit)** needs a dated, inspectable official instrument meeting the layer's authority, scope and content gate. **P (partial)** is relevant official evidence that fails at least one gate; record a reason code: *A* (not adopted), *S* (wrong/narrow scope), *C* (insufficient AI-use controls), or *V* (unresolved version/date). **U (unlocated)** means no qualifying document was found by the recorded search, *not* that no internal rule exists. Reason codes organize uncertainty; they were not retrospectively recoded for all 57 cases.

TABLE I
A short rulebook for deciding whether evidence is explicit

Level	To record E, the official document must show:
L1	Adopted or authoritative *national AI direction*.
L2	An inspectable *cross-sector operational* AI control, standard, regulation or guideline.
L3	*HE-applicable sector guidance* that actually specifies AI-use controls (not only readiness metrics).
L4	*At least one university/college rule* with institutional authority and applicable AI-use provisions.

The L3/L4 thresholds answer different questions. L3 concerns the existence of an HE-sector instrument; L4 asks only whether *one institution* has an applicable rule. L4=E does not mean that most universities have a policy. Conversely, the absence of a qualifying public L3 document does not invalidate an L4 rule. The mismatch is a comparison of two

different, explicitly stated *document-availability propositions*, not an estimate of sector compliance.

B. Availability, documentary links and real use are separate

Do not conflate three propositions: *document availability*, *an explicit inter-instrument link* (named citation/delegation in a clause), and *observed implementation*. **Only document-layer states were coded**; no edges or actual practice were measured. Thus E/E/E/E does not prove a transmitted policy chain.

C. What the 23 July 2026 cutoff does and does not prove

The primary frame uses instruments *dated or operative no later than* 23 July 2026, including some located retrospectively. For each record, keep three distinct dates: *issue/effective date*, *first verifiable public-posting date*, and *research retrieval date*. A pre-cutoff issue date alone does not prove public web availability at the cutoff. Unless a contemporaneous posting or archived capture verifies that fact, label it *historical-public-availability unverified*. The published counts are conditional on the retained document and coder states, *not* an independent snapshot of the web on 23 July.

III. How to Apply the Method

Figure 1 is a reusable procedure for a country, education system or group of institutions. **Step 1: freeze** the decision question, document cutoff, official institutional frame, language scope and exact search terms *before* seeing the outcome. **Step 2: locate** government, regulator, ministry and university texts, using the original document rather than a search snippet; record issuer, title, version, date, language, URL and pinpoint clause. **Step 3: code** each layer using Table I; keep any explicit cross-document citation in another field and never fill gaps by inference. **Step 4: report** all E/P/U states, missing evidence, sensitivity to disputed boundaries and raw source records. The reusable blank log and bilingual query templates accompany the PDF; they are *prospective templates*, not a claim that every original search string was historically retained.

For a U closure, search Arabic and English on the institution's domain, policy library, assessment pages, handbook and integrity rules, recording every completed surface, query and date; a later discovery is logged separately. A ministry announcement that guidance is under development can be P; it is not automatically an enacted HE control. If Arabic and English editions disagree, record the scope, controlling-language clause and version authority and flag the classification for adjudication, rather than silently choosing a convenient translation. This is especially relevant where an English text is identified as a translation of an Arabic original [14].

A. Worked example: how the same country can have E, E, P, E

The UAE provides a concise worked classification under the fixed cutoff. Its Outcome-Based Evaluation Framework (OBEF) v11.5 is an L3 candidate:

Layer	Located official instrument / decision reason	State
L1	UAE AI Strategy 2031: national direction [15]	E
L2	UAE AI Charter: cross-sector principles [16]	E
L3	OBEF v11.5: AI-readiness metrics, not operative use controls [17]	P
L4	UAEU student GenAI use policy: a university rule [18]	E

The profile is **E/E/P/E**. It says what is publicly *documentable*, not that the university policy was caused by the evaluation framework. Under an explicitly broader *evaluation-framework* threshold, UAE L3 changes to E; this decision is reported separately as sensitivity, never as a silent recode. The classification was not independently adjudicated by two coders.

IV. The GCC Evidence Frame

The fixed source date is **23 July 2026**. The retained 36-record register comprises seven L1, eight L2, five L3, eleven L4, and five GCC-regional records. Nineteen records have an Arabic and English location, fifteen an English location only, and two an Arabic location only. The register's document titles, issuer, date, locator and classification rationale are available online, but the full set of verbatim excerpts and independent archival hashes has *not* been reverified for this revision. Source recovery is not retrospectively presented as a fresh complete search.

The jurisdictional L4 finding requires only one qualifying university. We added a *separate, nonrandom corroboration panel*: officially recognized or licensed civilian higher-education institutions (HEIs), grouped by G1 public/general, G2 private/non-government and G3 public applied/specialized. The declared oldest-four-with-ties rule was applied inside each country–stratum cell, retaining smaller cells and excluding author-affiliated institutions before AI-policy search. The founding year is a selection variable, not a maturity score; selecting older institutions can favor better preserved web archives and limits generalizability.

Two eligibility corrections changed the selected members without changing that rule: a historical Bahrain regulator source placed Kingdom University in its eligible cell [19], [20]; UAE accreditation evidence established Emirates Institute for Banking and Financial Studies (EIBFS) as an eligible older private institution, displacing two later cases [21], [22]. The corrected panel is **57 institutions**. Importantly, the complete original pre-outcome eligible universe was not retained. The 57 are a *retrospectively corrected selected panel*, not a reconstructed fully frozen sampling frame. The companion workbook distinguishes known selection evidence from blank prospective full-frame forms; it is not a reconstructed original frame.

V. Results: What the Documents Show

Table II makes the main result visible without a color-coded country score. National AI direction is E in four states,

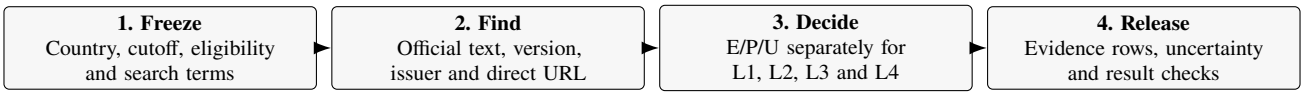


Fig. 1. Four-step *research workflow*. Arrows show audit steps, **not** causal transmission from national strategy to university policy. Black-and-white printing preserves all labels.

TABLE II
GCC RESULTS AT THE FIXED CUTOFF: FOUR GOVERNANCE LAYERS AND SEPARATE 57-INSTITUTION CORROBORATION COUNTS. THE LAST THREE COLUMNS ARE *SELECTED-CASE COUNTS*, NOT NATIONAL PREVALENCE RATES.

Jurisdiction	L1	L2	L3	L4	E	P	U	How to interpret
United Arab Emirates	E	E	P	E	5	3	5	HE OBEF is a sector evaluation framework, not an L3 control under primary gate.
Saudi Arabia	E	E	E	E	3	4	4	Four gates have visible instruments; no transfer claim is made.
Qatar	E	E	P	E	2	2	6	University rules visible despite partial L3 evidence.
Oman	E	E	U	E	2	1	5	Sultan Qaboos guide edition unverified; Majan handbook corroborates L4 [23], [24].
Bahrain	P	E	U	E	6	0	1	School-oriented initiative does not qualify as HE-specific L3 [25].
Kuwait	P	P	U	E	2	2	4	Kuwait University rule provides an L4 example.
Total (selected HEIs)					**20**	**12**	**25**	**57 cases across six states**.

operational governance in five, HE-sector controls in one and an institution-level rule in all six. In *five of six states*, a visible L4 rule coexists with an L3 state other than E. Define the descriptive count

$$D = \#\{j : L_{4j} = \mathsf{E} \text{ and } L_{3j} \neq \mathsf{E}\} = 5. \tag{1}$$

A higher level's partial or unlocated public documentation cannot be interpreted as proof of a broken regulation or of institutional non-compliance. The 57-case panel contains **20 E, 12 P and 25 U**; each country has at least two E and at least one non-E case. They show observable diversity within a *purposefully selected panel*; the fractions 20/57, 12/57 and 25/57 are *not representative population estimates*. The full 57-row links remain in the supplement instead of a micro-font table.

A. What changes when a source or rule changes?

TABLE III
EXPLICITLY SCOPED SENSITIVITIES; D IS A COUNT, DISPLAYED AS D/N

Scope / scenario	L3 E	L4 E	D/N
Primary 36-record chain	1/6	6/6	5/6
Remove SQU *only from 36-record chain*	1/6	5/6	4/6
Include Majan L4 (cross-panel fallback)	1/6	6/6	5/6
Broader UAE OBEF scope	2/6	6/6	4/6
Leave out Saudi Arabia	0/5	5/5	5/5
Leave out any other state	1/5	5/5	4/5

Scope note: Majan is included from the separate 57-institution panel, not the 36-record-only deletion scenario. Its E label is based on the 2025–26 institutional handbook, Sec. 5.7.7 [24], [26].

The Oman diagnostic needs two scopes. Removing Sultan Qaboos University (SQU) from the *36-record register only* makes that register's Oman L4 unlocated; it does *not* eliminate a second qualifying Oman L4 source in the *separate 57-institution panel*. Majan University College's 2025–26 official handbook §5.7.7 gives institution-wide student assessment rules, including permitted study support, restrictions on submitted work and disclosure, and the college announced the handbook publicly on 20 October 2025 [24], [26]. Retaining Majan as an L4 fallback keeps Oman E. These calculations test different source universes; neither estimates implementation. Broadening the UAE OBEF gate reduces D to four by changing a *sector* judgment [17].

A brief P-only interval representation is reported in the companion guide in the context of interval-valued fuzzy sets [27], [28] and earlier work on fully fuzzy linear systems and fuzzy matrix equations [29], [30], [31], [32], [33]: $F_3 = [1/6, 3/6]$, $F_4 = [1, 1]$, minimum difference $1/2$. The latter studies develop algebraic solution methods, whereas the present intervals retain partial documentary evidence without forcing a single crisp value. These bounds are algebra from the categorical states, not measured coder uncertainty, missing-source coverage, or confidence intervals.

B. Pinpoint checks on decisive public texts

The retained Saudi L3=E record SAU03 cites *Generative AI in Education* from the Saudi Data and Artificial Intelligence Authority (SDAIA) as 2024, while its public catalog dates a similarly titled entry 19 Nov. 2023 [34]. Edition identity and the exact decisive passage remain unverified. A second official instrument, the National eLearning Center's *Regulations and Guidelines for the Use of GenAI Tools and Applications*

in Digital Learning (v1.0), defines educational institutions to include universities/colleges (p. 3) and gives explicit institutional, assessment and disclosure controls (pp. 4–7) [35]. Its release is reported in the center's 2024 annual report [36]. This provides corroboration, not a new entry in the fixed 36-record register; the SAU03 edition and decisive passage remain unverified.

For a second bilingual authority example, Qatar University's presidential instructions were approved and effective 25 Dec. 2025. Article 3 specifies disclosure and related controls (pp. 3–4); the official English edition expressly gives precedence to Arabic if text differs [14]. These pinpoint citations illustrate an E decision; they do not establish independent double coding. SQU's official policy index lists a GenAI teaching and assessment guide, but its precise authoritative version/effective date still needs separate source closure [23].

VI. A Reusable Evidence-to-Decision Certificate

The 4L-PEA procedure is most useful when its *unit of reuse* is not a country color or an E/P/U label, but a short certificate that another analyst can falsify. We specify the following **minimum admissible certificate** for every decisive layer classification. It is a *prospective reporting schema*, not a claim that all historic GCC cases already have complete certificates.

A reader can replay one decision as follows. First, locate the cited official instrument and its numbered provision. Second, identify the issuer and affected audience. Third, apply *only* the gate in Table I, recording why any official evidence remains P. Fourth, preserve the passage, retrieval date and source edition so an independent coder can reach or challenge the same state. A decision with a direct URL but no operative passage remains *traceable in principle*, not fully reverified.

Two boundary examples make the gates testable. A ministry's higher-education evaluation guide that asks whether universities are ready to integrate AI, without prescribing assessment, data, disclosure or oversight controls, is *P–C at L3* under the primary gate (the UAE OBEF case). A college handbook with an adopted, college-wide assessment provision allowing some forms of AI assistance while restricting submitted work and requiring disclosure is *E at L4* when its authority, audience and date are verified (the Majan case). Neither classification says that a student's behavior complied with the instrument; nor does the second prove a sector-wide policy. These are different *documentary propositions*, not consecutive stages of a measured transmission process.

Predeclared challenge checks. A future replication should (a) ask a second coder, blinded to the first decision, to assign E/P/U and P reason codes from the same frozen text; (b) test whether reclassifying an L3 boundary record changes the descriptive *D* count; (c) distinguish *dropping one source in the 36-record register* from *withdrawing all qualifying university-level records* in the separate 57-case panel; and (d) test the historical web-availability flag against a timestamped official release or archived capture. These tests are *instructions for validation*, not validations performed by this study.

The *transferable output* is therefore a **source–gate–decision–limitation tuple**: an exact passage establishes an admissible documentary claim under a stated rule, while a separate field states what the evidence cannot establish. This supports cumulative research across institutions and jurisdictions without turning ordinal-looking E/P/U symbols into a country quality ranking or treating arithmetic reproduction as independent source verification.

VII. What Another Researcher Can Reuse

For a fully replayable audit, a reader must move from each claim back to its governing clause, not just its homepage. Every new audit should release: **(i)** a source register with issuer, edition, issue/effective date, language, document URL, exact clause and layer-specific decision; **(ii)** an eligibility list showing institutions included, excluded and tied at the boundary; and **(iii)** an E/P/U decision sheet with search/closure dates, explicit uncertainty flags and code that recalculates tables. An open hyperlink alone cannot substitute for the controlling provision or its date. A source URL that merely records an unsuccessful U search must be labelled as a *search locator*, not cited as a positive policy.

For university staff, we propose *eight public-rule questions*: who approved the operative version; whom it covers; what AI use is allowed in assessments; when disclosure is required; which tools or data are restricted; who verifies outputs; how fairness and appeals work; and who maintains future revisions. These fields address recurring traceability questions and align with general UNESCO/OECD governance concerns [37], [38]. They are a *documentation proposal*, not a validated policy-quality score. The companion field guide and editable worksheet turn each question into a source-linked check with blank evidence cells.

Supplementary materials and code. The **field guide and 57-institution linked appendix (PDF)** is available with the **version-matched research archive** (57-case CSV, six-state matrix, provisional 36-source register, blank audit template and result checks). This is an author-linked archive; the supplement includes prospective templates and selected records, but not a completed independent replay.

VIII. Limits and Conclusion

The results concern *public document availability at a declared cutoff*. They do not measure instrument-to-instrument transmission, implementation effectiveness, policy quality, causation, regional prevalence or missing-source rates. The historical search log is incomplete for exact independent replay; the 36-source register is provisional and only selected decisive-source checks have been freshly corroborated. The 57-case membership remains subject to a full original-frame

TABLE IV

REUSABLE 4L-PEA DECISION CERTIFICATE: MINIMUM FIELDS, ACCEPTANCE TESTS AND FAILURE HANDLING (PROSPECTIVE PROTOCOL; NOT A RETROSPECTIVELY COMPLETED DATASET).

Certificate field	What a second reader should be able to inspect	If the field is missing or disputed
Document identity	Issuer, official title, document ID, version, language and direct persistent locator.	Mark identity unresolved; never substitute an institutional homepage for a governing clause.
Time provenance	Issue date, effective date, evidence of first public availability and date retrieved.	Report availability at cutoff as unverified; do not infer historic visibility from a current URL.
Authority and audience	Approving body and the persons/activities governed, with an exact scope clause.	Use P–A or P–S where official evidence exists but fails adoption or scope.
Operative provision	Page/section and bounded quotation that shows the specific L1, L2, L3 or L4 gate.	Use P–C if AI-related but nonoperative; do not upgrade general AI advocacy to an assessment rule.
Search and decision trace	Arabic/English queries, official-site surfaces, stop date, E/P/U decision, reason code and adjudication status.	Use U only with documented search closure; otherwise record unresolved rather than policy absence.
Audit status	Independent coder decision (if actually obtained), disagreement disposition and source-archive reference/hash.	State explicitly "not independently verified" where this evidence does not exist.

completeness check. The E/P/U states were not independently double-coded, so no inter-rater agreement or Arabic–English adjudication rate is claimed. A future verification round should freeze source copies, use two independent coders, report a confusion matrix and agreement, adjudicate conflicts with controlling passages and record any revised outcomes before publication. These are *outstanding validations*, not completed experiments.

The result conditional on the frozen coding is simple: at least one documented university-level rule is identified in each GCC jurisdiction, while the distinct HE-applicable sector control gate is E in one (two under a broader UAE scope). The portable finding is the *difference between visible evidence, explicit linkage and actual practice*, together with a reusable source-to-claim workflow. The accessible ledger and companion workbook make the *arithmetic and future audit procedure* reusable; full historical source recovery and independent coding remain outstanding.

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